Your grade: 100%

Your latest: **100%** • Your highest: **100%** • To pass you need at least 80%. We keep your latest score.

Next item

Instructions

1. Which of the following steps are involved in the analytic development process for training AI models? (Select all that apply) 1 / 1 point

- [x] Data collection

  Nice work

  Correct! Data collection is one of the initial steps in the analytic development process.

- [x] Model validation

  Nice work

  Correct! Model validation is a critical step in the analytic development process.

- [ ] User interface design
- [ ] System maintenance
- [x] Hyperparameter tuning

  Nice work

  Correct! Hyperparameter tuning is involved in refining the model's performance during the analytic development process.

2. What is the primary purpose of the analytic development process in AI model training? 1 / 1 point

- ( ) To develop a robust testing protocol
- ( ) To continuously monitor model performance
- ( ) To collect and pre-process data
- (•) To systematically design, build, and test models

  Nice work

  Correct! The primary purpose of the analytic development process is to systematically design, build, and test AI models, ensuring they meet defined requirements.

3. What does the ROC Curve plot in the context of evaluating AI models? 1 / 1 point

- ( ) Sensitivity vs. Specificity
- ( ) Accuracy vs. Error Rate
- (•) True Positive Rate vs. False Positive Rate
- ( ) Precision vs. Recall

  Nice work

  Correct! The ROC Curve plots the True Positive Rate against the False Positive Rate.

4. You are developing an AI model to detect fraudulent transactions. During the analytic development process, you realize that your model's performance varies significantly based on the dataset's quality. What steps would you prioritize to improve your model? 1 / 1 point

- [x] Hyperparameter tuning

  Nice work

  Correct! Tuning hyperparameters can help optimize the model's performance based on the given data.

- [x] Data collection

  Nice work

  Correct! Collecting high-quality, relevant data is crucial to improving model performance.

- [x] Model validation

  Nice work

  Correct! Model validation helps ensure the model performs well on different datasets and generalizes properly.

- [ ] System maintenance

5. Which metric is used to evaluate the overall performance of a binary classifier by considering both the true positive rate and false positive rate? 1 / 1 point

- (•) AUC
- ( ) Precision-Recall Curve
- ( ) Confusion Matrix
- ( ) ROC Curve

  Nice work

  Correct! The Area Under the ROC Curve (AUC) is a metric used to evaluate the overall performance of a binary classifier.